LDE3

Global star-hot Jupiter magnetic interactions : implications for upper atmosphere heating

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Context

Exploring the diversity of planetary atmospheres calls for sophisticated models that combine photochemistry, atmospheric dynamics, and interactions with the host star. To investigate **magnetic coupling** in exoplanetary atmospheres, we need tighter constraints on their **magnetic fields**. Yet, not a single exoplanetary magnetic field has been unambiguously characterised at the moment.

➢ We model stellar winds of a close-in exoplanet system to investigate conditions for **Star-Planet Magnetic Interactions** (SPMI) and their potential as an indirect probe of **exoplanetary magnetic fields**. We also present preliminary estimates of the **Ohmic heating** in the ionosphere.

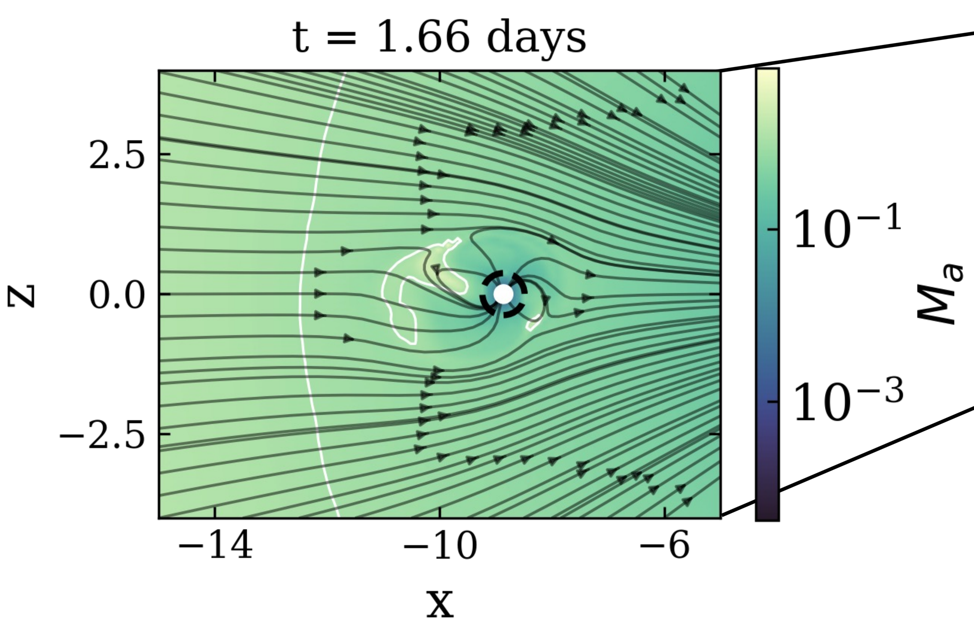
SPMI model

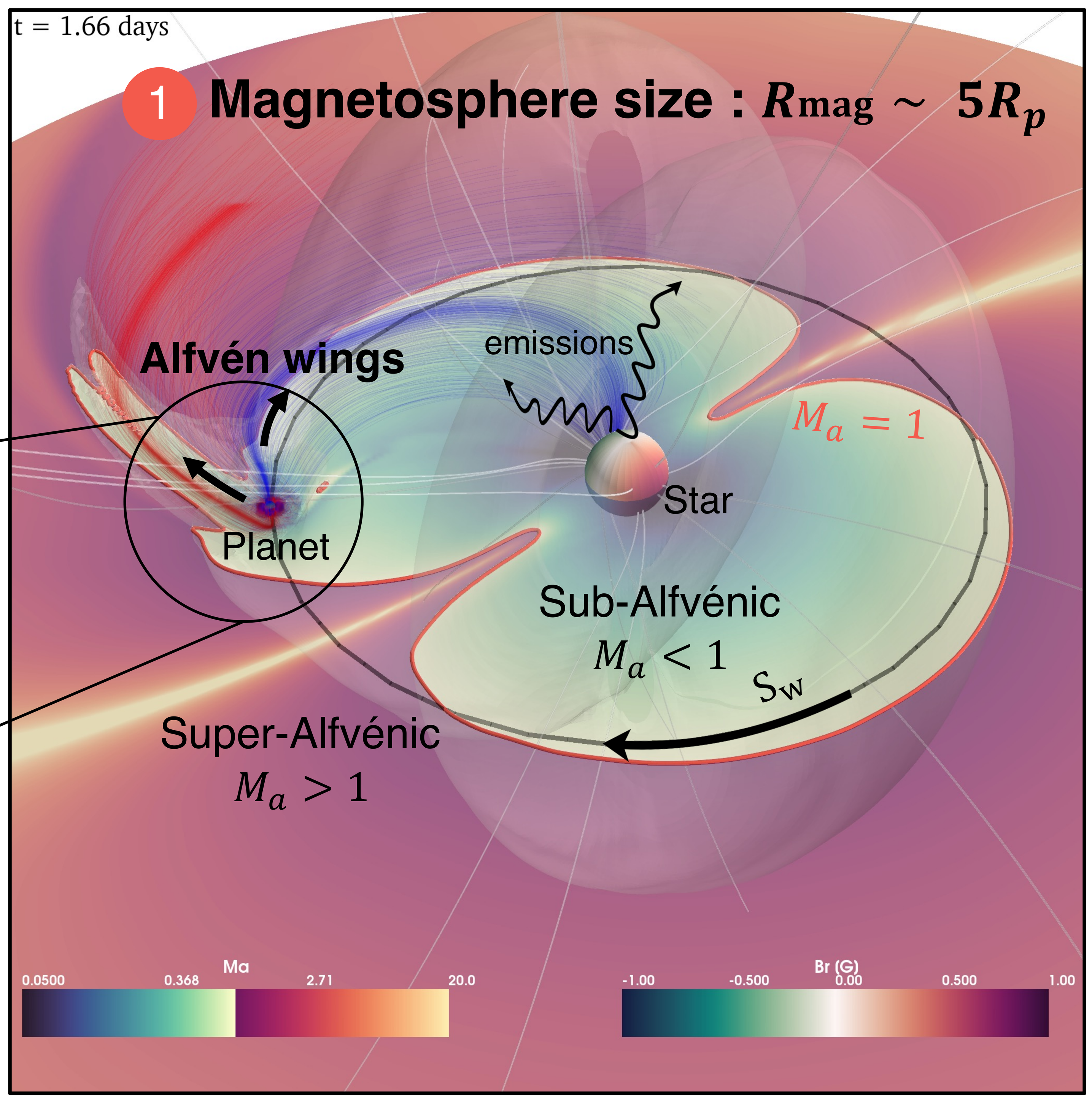
- Need to **model the 3D environment** of the star : use of PLUTO[1] to resolve MHD equations in order to characterise stellar winds.
- WindPredict[2] model which includes the planet **self-consistently** and an implementation of the **rotating frame,** allowing the planet and the star to be fixed in the grid while properly solving for the orbital motion and stellar rotation.
- The **magnetosphere** of the planet is modelled as a simple dipole. Inside the internal boundary, the MHD equations don't evolve with time.

$$M_a = \frac{v_{wind}}{v_a}$$

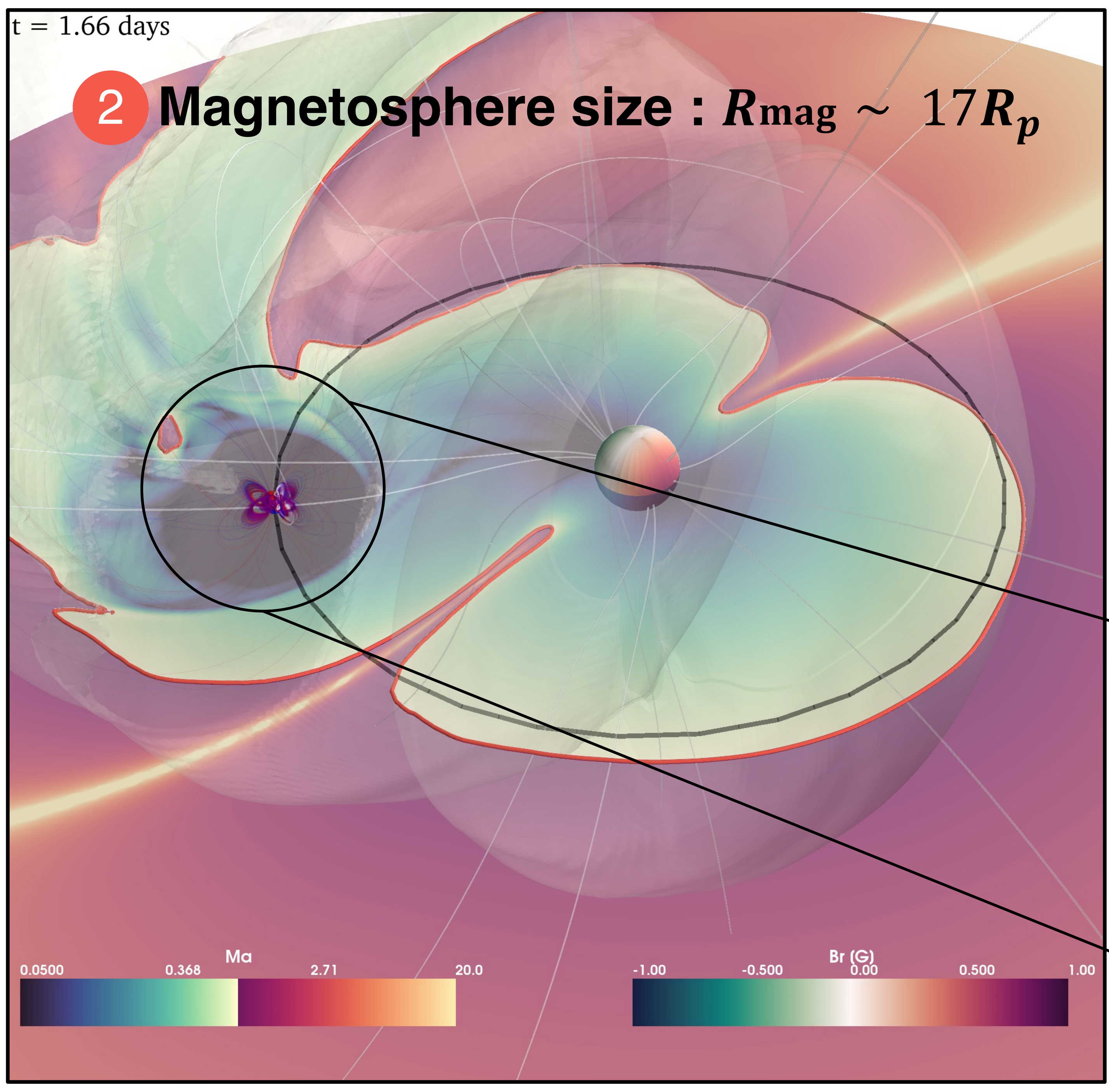
$$S_w = \frac{1}{\mu_0} \boldsymbol{E}\times\boldsymbol{B}$$

When sufficiently close, an exoplanet can channel energy towards its host star by exciting Alfvén waves that propagate along the so-called Alfvén wings.



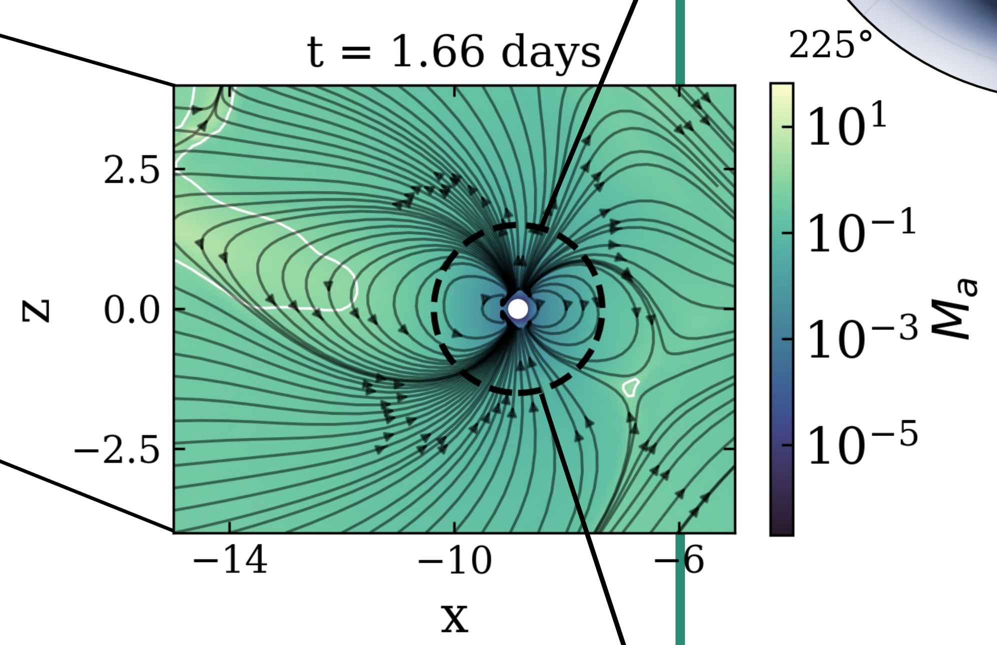


Same model, but stronger magnetosphere

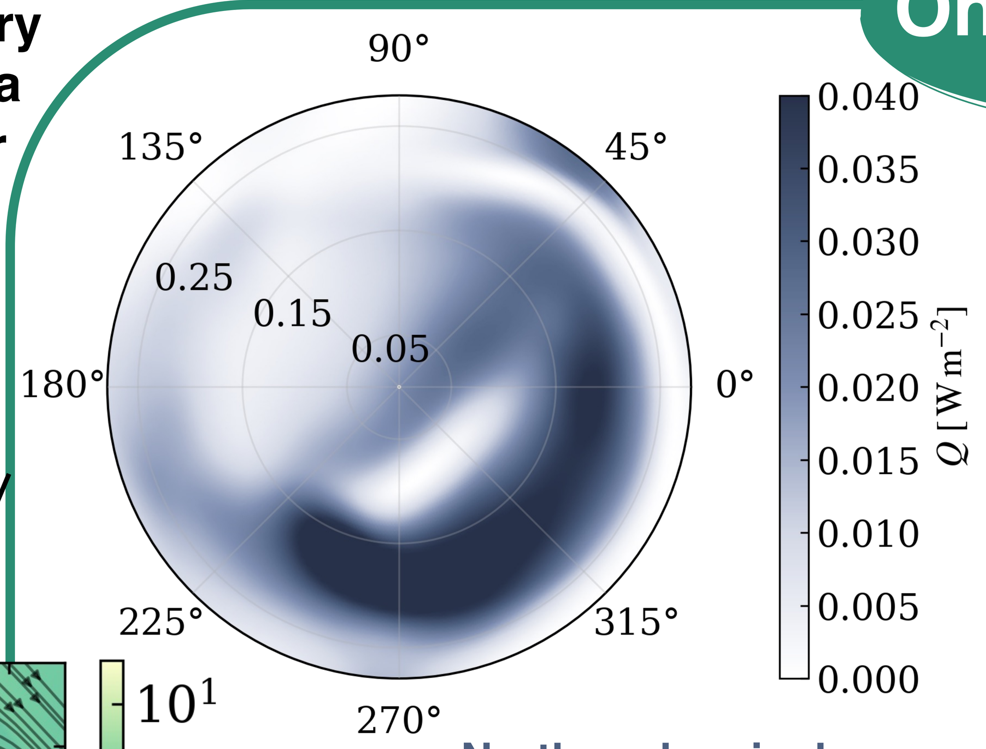


For realistic magnetic fields of hot Jupiters, the planetary magnetosphere occupies a large portion of the stellar corona.

Legend
Internal boundary
Planet



Ohmic heating



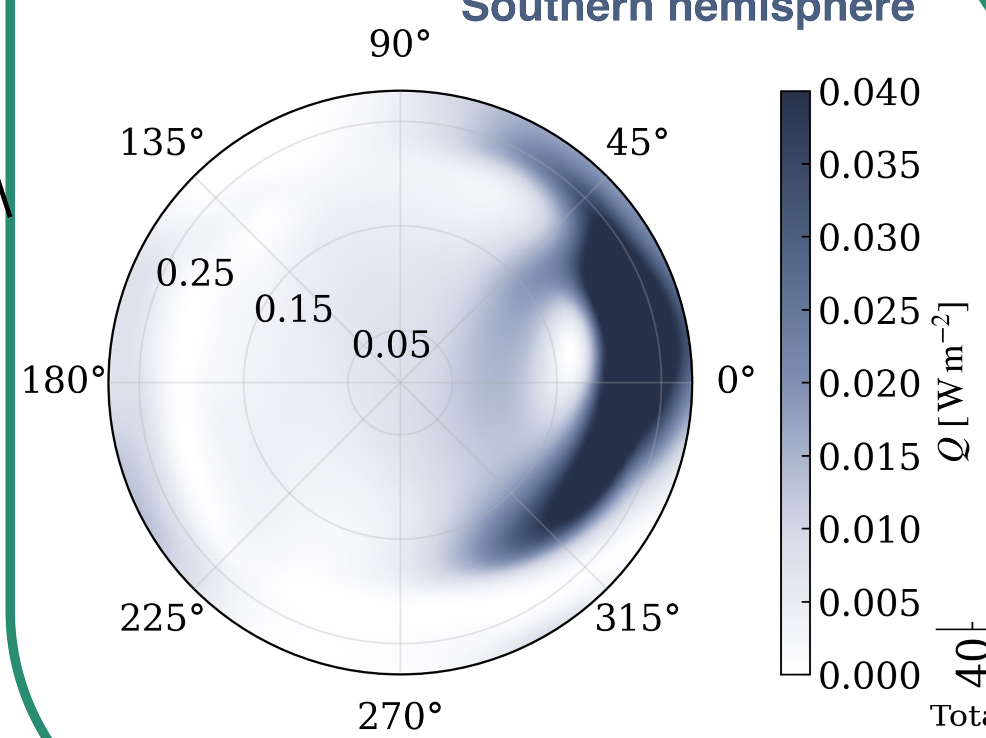
Northern hemisphere

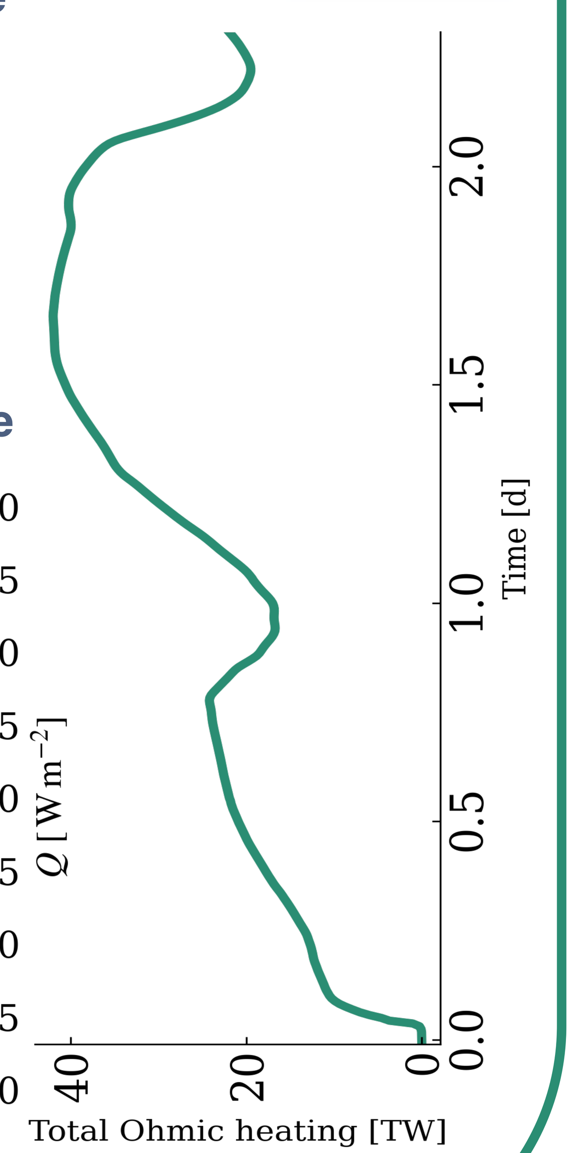
NH 3D movie

SH 3D movie

Choice of a uniform Pedersen conductivity of 10^4 S[3]

Southern hemisphere





Preliminary results

Conclusion

✓ We modelled the **stellar environment of a close-in exoplanet and its magnetosphere**. Novel rotating-frame approach: the astrosphere evolves while the planet remains fixed in the grid.

✓ We present preliminary estimates of the **total Ohmic heat** deposited into the ionosphere of the close-in exoplanet for the large magnetosphere case.

✓ The **connectivity is modulated** by the complex **magnetic topology** of the star as well as the intensity of the exoplanet's magnetosphere

Future Directions

❑ Compare the model with observations of SPMI[4,5,6]
❑ Better estimation of Ohmic heating (e.g. non-uniform Pedersen conductivity)
❑ Explore the impact of different magnetosphere sizes.

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